# Minimal Randomized Benchmarking and Interleaved RB with Leakage, SPAM, and Coherent Error

Dan Hawkley

April 11, 2026

**Abstract**

This note presents a minimal benchmarking framework for randomized benchmarking (RB) and interleaved randomized benchmarking (IRB), with explicit modeling of leakage, state-preparation and measurement (SPAM) error, and coherent error. The framework progresses from baseline RB to SPAM-aware fitting, then to interleaved target-gate characterization, and finally to a `pytket`-backed workflow using real circuit objects. The resulting pipeline is small enough to inspect end-to-end while still demonstrating both the usefulness and the limitations of standard RB assumptions. This compact framework is intended as a reproducible reference implementation connecting benchmarking theory to circuit-level workflows.

## 1 Overview

Randomized benchmarking (RB) is a standard method for estimating average gate quality from return probability as a function of sequence length [1, 2]. Interleaved randomized benchmarking (IRB) extends this idea by inserting a fixed target gate between random gates, allowing additional decay to be associated with that gate [3]. In idealized settings the observed decay is often fit by

$$P(m) = Ap^m + B, \tag{1}$$

where $m$ is sequence length, $p$ is the decay parameter, and $A, B$ absorb state-preparation and measurement (SPAM) effects.

This project implements a minimal version of that workflow with four progressively richer ingredients:

1. baseline RB,
2. SPAM-aware fitting,
3. interleaved target-gate characterization,
4. coherent target-gate bias inside a `pytket`-backed circuit workflow.

Repository: github.com/thinkthoughts/ion-benchmark-lab
Colab notebook: Open in Colab

## 2 Method

The final notebook uses one-qubit `pytket.Circuit` objects built from a small self-inverse gate alphabet. Reference RB sequences are sampled at random, and IRB sequences are formed by inserting a fixed target gate between those random gates. The inverse is then constructed by reversing the sequence so that the noiseless circuit returns to the reference state.

Noise is introduced through four mechanisms:

- **stochastic error**: random Pauli-like gate disturbance,
- **leakage**: absorbing failure channel,
- **SPAM**: imperfect preparation and readout,
- **coherent target-gate bias**: a systematic over-rotation applied to the interleaved target gate.

Monte Carlo simulation estimates return probability over many random sequences and shots per sequence. Each dataset is then fit to the RB-style model

$$P(m) = Ap^m + B. \tag{2}$$

## 3 Main Result

Figure 1 shows the flagship comparison among reference RB, interleaved RB, and interleaved RB with coherent target-gate bias.

The qualitative pattern is the expected one:

- reference RB decays slowest,
- IRB decays faster because the target gate contributes extra error,
- coherent target-gate bias introduces additional distortion beyond purely stochastic interleaving.

This makes the notebook useful in two directions at once: as a compact demonstration of standard RB/IRB logic, and as a minimal example of how coherent error can complicate the interpretation of interleaved decay parameters. Coherent errors are especially relevant because RB-style decay models are most reliable when noise is effectively stochastic [5, 6].

## 4 Why this repo is useful

The main value of the repo is not scale; it is clarity. The workflow is small enough that every assumption is visible, yet rich enough to show:

- why RB works well under effectively stochastic noise,
- how IRB isolates target-gate contributions,
- how leakage and SPAM affect measured return probability,

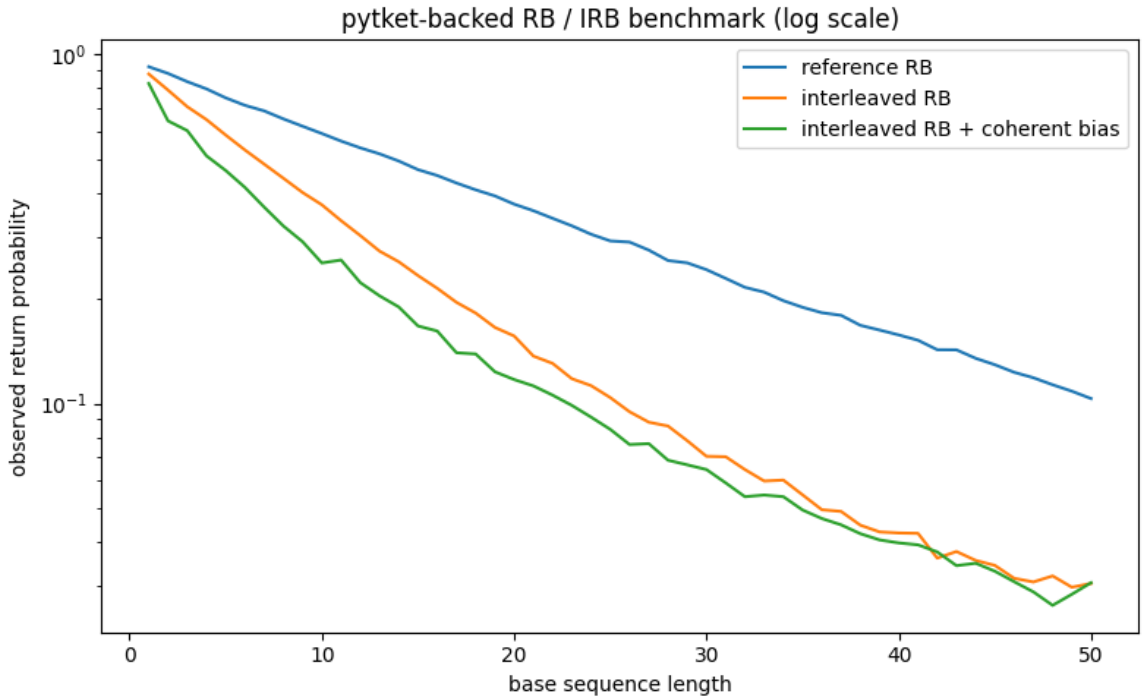


Figure 1: Reference RB vs interleaved RB vs interleaved RB with coherent target-gate bias. Interleaving increases decay, and coherent target-gate bias pushes decay further while perturbing the fitted behavior.

- why coherent bias can weaken simple exponential interpretations.

Because the final version is implemented with `pytket`, the project also connects naturally to hardware-facing circuit workflows rather than remaining purely abstract.

## 5 Files

The main public artifact is:

```
notebooks/01_leakage_benchmark_v0_9.ipynb
```

Earlier development notebooks can remain available in:

```
notebooks/history/
```

## 6 Conclusion

This project provides a compact, reproducible benchmarking pipeline that moves from baseline randomized benchmarking to interleaved target-gate characterization with leakage, SPAM, and coherent bias. As a repo, notebook, and paper bundle, it is intended to be easy to inspect, run, and extend.

## References


[1] Easwar Magesan, Jay M. Gambetta, and Joseph Emerson. Scalable and Robust Randomized Benchmarking of Quantum Processes. *Physical Review Letters*, 106(18):180504, 2011.

[2] Easwar Magesan, Jay M. Gambetta, and Joseph Emerson. Characterizing Quantum Gates via Randomized Benchmarking. *Physical Review A*, 85(4):042311, 2012.

[3] Easwar Magesan, Jay M. Gambetta, and Joseph Emerson. Efficient Measurement of Quantum Gate Error by Interleaved Randomized Benchmarking. *Physical Review Letters*, 109(8):080505, 2012.

[4] Jay M. Gambetta, Jerry M. Chow, and Matthias Steffen. Characterization of Addressability by Simultaneous Randomized Benchmarking. *Physical Review Letters*, 109(24):240504, 2012.

[5] Joel J. Wallman and Joseph Emerson. Estimating the Coherence of Noise. *New Journal of Physics*, 17(11):113020, 2015.

[6] Timothy J. Proctor et al. What Randomized Benchmarking Actually Measures. *Physical Review Letters*, 119(13):130502, 2017.